

Physics Olympiad L2-21

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Problem 1: The Tangential-Polar Coordinate System

A two-dimensional coordinate system useful for orbit problems is the tangential-polar coordinate system. In this system, the position of a point P on a curve is described by two quantities:

- The distance r from a fixed point O to a point P on the curve;
- The perpendicular distance p from O to the tangent to the curve at P ;

where the point O is taken as the origin of the coordinate system.

Consider two neighbouring points P and Q on the curve given by $r(p)$. The normals to the curve at points P and Q intersect at C , which, in the limit $Q \rightarrow P$, is called the centre of the osculating circle at P . Its radius $CP = \rho$ is called the radius of curvature at P .

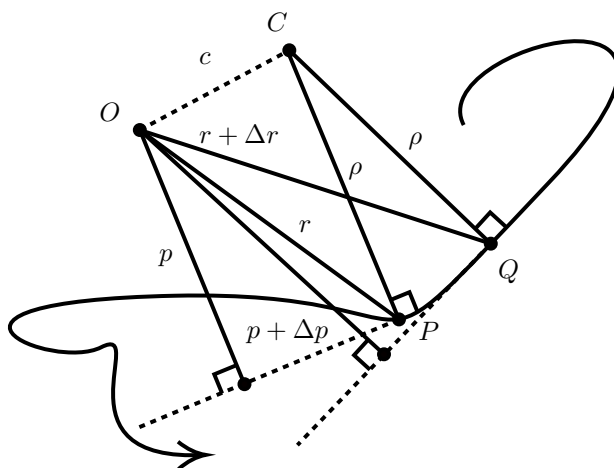
- Obtain an expression for the radius of curvature ρ in terms of r and dr/dp ;

Consider a particle of mass m moving under the influence of an attractive central force of magnitude f , directed towards the origin O . Let the speed be given by $v(r)$. Then:

- Find an expression for the magnitude of the force in terms of m , v and dv/dr ;
- Find an expression for the kinetic energy in terms of f , p , r and dp/dr ;
- Show that $h = mpv$ is a conserved quantity;
- Show that the force can be expressed entirely in terms of h , m , r , p and dp/dr ;

Let φ denote the angle between the tangent to the curve at P and the radius vector OP . Determine the central force law under which a particle must move so as to describe the following trajectories:

- A circle of radius a that is centered a distance c away from the force origin;
- A centered equiangular spiral, for which the angle φ is constant along the curve;



Problem 2: Emission From An Inelastic Collision

A hydrogen atom moving with kinetic energy E_k collides head-on with an identical hydrogen atom initially at rest. Both atoms are initially in their ground states. Assume that the collision is inelastic and that it is sufficiently short that photon emission during the collision itself can be neglected.

- Estimate the kinetic energy E_k^{rel} at which the motion of a hydrogen atom becomes relativistic;
- Determine the minimum kinetic energy E_k^{min} of the incident atom for which the collision can leave one of the atoms capable of emitting a photon. Find the wavelength λ at which it would be emitted;
- Determine the velocities of the two atoms immediately after the collision at the threshold energy E_k^{min} ;

Problem 3: Rotating Crystal

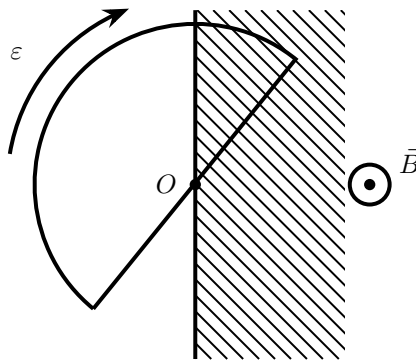
A monochromatic beam of X -rays with wavelength $\lambda = 174 \text{ pm}$ falls on the surface of a crystal rotating about an axis parallel to its surface and perpendicular to the direction of the incident beam. As the crystal rotates, diffraction maxima are observed for different orientations of the crystal. The directions of the second- and third-order diffraction maxima from the same family of lattice planes form an angle $\varphi = 60^\circ$. Find the corresponding interplanar distance.

Problem 4: Rotating Loop

A wire loop enclosing a semi-circle of radius a is located on the boundary of a uniform magnetic field of induction $\vec{B}$. At the moment $t = 0$ the loop is set into rotation with a constant angular acceleration $\vec{\varepsilon}$ about an axis O coinciding with a line of vector $\vec{B}$ on the boundary.

- Find the emf induced in the loop as a function of time t ;
- Draw the approximate plot of this function;

The arrow in the figure shows the direction of rotation and the emf direction taken to be positive.

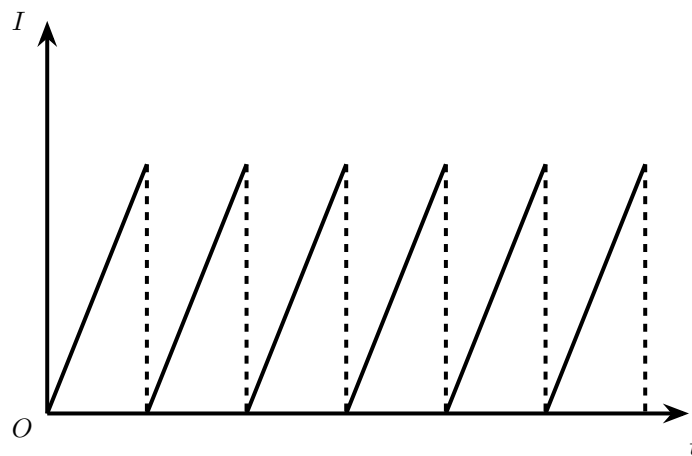
**Problem 5: Rotating Rod**

A smooth uniform rod AB of mass M and length l rotates freely with an angular velocity ω_0 in a horizontal plane about a stationary vertical axis passing through its end A . A small sleeve of mass m starts sliding along the rod from the point A . Find the velocity v_r of the sleeve relative to the rod at the moment it reaches its other end B .

Problem 6: Effective Current

Find the effective value of current if its mean value is I_0 and its dependence is:

- as shown in the figure below;
- $I \sim |\sin(\omega t)|$



Appendix

Useful Mathematical Formulae and Physical Constants

Important: Infinite series that are marked with (*) are only valid under the condition that $|x| < 1$.

Taylor Expansions

$$\begin{aligned}\sin(x) &= x - \frac{x^3}{6} + \frac{x^5}{120} + \mathcal{O}(x^7) \\ \cos(x) &= 1 - \frac{x^2}{2} + \frac{x^4}{24} - \frac{x^6}{720} + \mathcal{O}(x^8) \\ \operatorname{tg}(x) &= x + \frac{x^3}{3} + \frac{2x^5}{15} + \mathcal{O}(x^7) \\ \arcsin(x) &= x + \frac{x^3}{6} + \frac{3x^5}{40} + \mathcal{O}(x^7) \\ \arctg(x) &= x - \frac{x^3}{3} + \frac{x^5}{5} + \mathcal{O}(x^7) \\ \exp(x) &= 1 + x + \frac{x^2}{2} + \mathcal{O}(x^3) \\ \ln(1+x) &= x - \frac{1}{2}x^2 + \mathcal{O}(x^3) \quad (*) \\ (1-x)^{-1} &= 1 + x + x^2 + \mathcal{O}(x^3) \quad (*) \\ (1+x)^{-1} &= 1 - x + x^2 + \mathcal{O}(x^3) \quad (*) \\ (1+x)^{+\frac{1}{2}} &= 1 + \frac{1}{2}x - \frac{1}{8}x^2 + \mathcal{O}(x^3) \quad (*) \\ (1+x)^{-\frac{1}{2}} &= 1 - \frac{1}{2}x + \frac{3}{8}x^2 + \mathcal{O}(x^3) \quad (*) \\ (1+x)^{+\frac{3}{2}} &= 1 + \frac{3}{2}x + \frac{3}{8}x^2 + \mathcal{O}(x^3) \quad (*) \\ (1+x)^{-\frac{3}{2}} &= 1 - \frac{3}{2}x + \frac{15}{8}x^2 + \mathcal{O}(x^3) \quad (*)\end{aligned}$$

Inverse Hyperbolic Functions

$$\begin{aligned}\operatorname{arsinh}(x) &= \ln(x + \sqrt{x^2 + 1}) \\ \operatorname{arcosh}(x) &= \ln(x + \sqrt{x^2 - 1}) \\ \operatorname{artgh}(x) &= \frac{1}{2} \ln \left(\frac{1+x}{1-x} \right) \\ \operatorname{arctgh}(x) &= \frac{1}{2} \ln \left(\frac{x+1}{x-1} \right)\end{aligned}$$

Indefinite Integrals

$$\begin{aligned}\int (1-x^2)^{-\frac{3}{2}} dx &= \frac{x}{\sqrt{1-x^2}} + C \\ \int (1+x^2)^{-\frac{3}{2}} dx &= \frac{x}{\sqrt{1+x^2}} + C \\ \int (1-x^2)^{-\frac{1}{2}} dx &= \arcsin(x) + C \\ \int (1+x^2)^{-\frac{1}{2}} dx &= \operatorname{arsinh}(x) + C \\ \int (1-x^2)^{+\frac{1}{2}} dx &= \frac{\arcsin(x) + x\sqrt{1-x^2}}{2} + C \\ \int (1+x^2)^{+\frac{1}{2}} dx &= \frac{\operatorname{arsinh}(x) + x\sqrt{1+x^2}}{2} + C\end{aligned}$$

Useful Physical Constants

Constant	Symbol	Value	Measuring Units
Light speed in vacuum	c	$3,000 \times 10^8$	m/s
Rest mass of electron	m_e	$9,109 \times 10^{-31}$	kg
Rest mass of proton	m_p	$1,673 \times 10^{-27}$	kg
Rest mass of neutron	m_n	$1,675 \times 10^{-27}$	kg
Rest mass of muon	m_μ	$1,884 \times 10^{-28}$	kg
Rest mass of tau-lepton	m_τ	$3,168 \times 10^{-27}$	kg
Elementary charge	e	$1,602 \times 10^{-19}$	C
Planck's constant	h	$6,626 \times 10^{-34}$	Js
Boltzmann's constant	k	$1,381 \times 10^{-23}$	J/K
Gravitational constant	γ	$6,673 \times 10^{-11}$	Nm ² /kg ²
Dielectric permittivity of vacuum	ε_0	$8,854 \times 10^{-12}$	F/m
Avogadro's number	N_A	$6,022 \times 10^{23}$	mol ⁻¹
Rydberg's constant	R_∞	$1,097 \times 10^7$	m ⁻¹
Magnetic permeability of vacuum	μ_0	$4\pi \times 10^{-7}$	H/m
Universal gas constant	R	8,314	J/(mol · K)